Skin Cancer Detection Using Deep Learning

Project Overview, Project Flow, and How the IEEE Paper Follows the Project

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1 What the Project Is About

In one sentence: we are building a deep-learning system that looks at a photograph of a skin lesion and tells a doctor whether it is likely to be skin cancer (melanoma, basal cell carcinoma, squamous cell carcinoma, or actinic keratosis) or a benign lesion. It also shows *where* in the image it looked, how confident it is, and whether the patient should be referred to a specialist.

1.1 Problem

- Skin cancer is common, and survival depends on detecting it early. There are few dermatologists, especially outside cities.
- Visual diagnosis is difficult even for experts. Many AI papers report 95–99% accuracy, but their results are often inflated by duplicate images and tested only on data similar to the training data. In real clinics, AI has shown low specificity and weaker results on darker skin.

1.2 Objectives

1. Classify skin lesion images into 8 classes (4 malignant or pre-malignant, 4 benign) and give a malignancy probability.
2. Build a **hybrid ensemble**: EfficientNetV2-S, ConvNeXt-T (CNNs) and Swin-T (transformer), guided by lesion segmentation and patient metadata (age, sex, body site).
3. Handle class imbalance (rare cancers) with focal loss and class weights.
4. Explain every prediction with a Grad-CAM heatmap.
5. Evaluate **honestly**: no data leakage, test on *external* datasets, and report results for each skin tone.
6. Compare the proposed model with standard algorithms and published results.

1.3 Scope

In scope	Out of scope
Dermoscopic and smartphone images of single lesions	Whole-body scanning; video
8-class diagnosis + malignant/benign referral decision	Treatment recommendation
Public, de-identified datasets	Collecting new patient data
Research prototype with a simple demo interface	Certified medical device; replacing doctors

2 System Flow (How the System Works)

Figure 1 shows what happens when a user (doctor or health worker) submits an image.

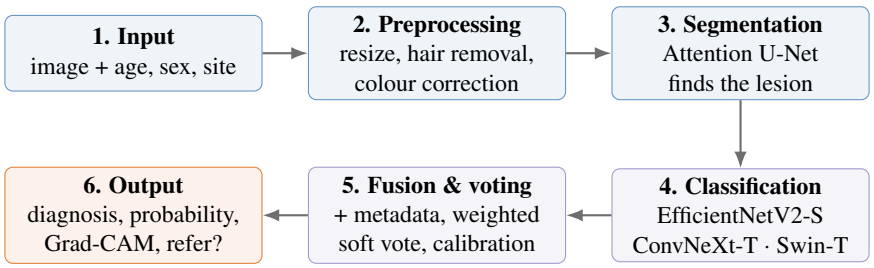

Figure 1: System flow at prediction time.

1. **Input.** A dermoscopic or smartphone photograph of the lesion, with optional age, sex, and body site.
2. **Preprocessing.** Resize to 224×224 (done). Removing hair and ruler marks and correcting colour between cameras are planned (next stage).
3. **Segmentation (next stage).** An Attention U-Net will outline the lesion and crop around it, so that the classifier focuses on the lesion, not the surrounding skin.
4. **Classification.** Three different networks each predict the 8 class probabilities. CNNs capture local texture and colour; the Swin transformer captures the global shape and symmetry.
5. **Fusion and voting (done).** Temperature scaling makes each model's probability trustworthy, and the three outputs are averaged (equal weights worked best). Patient metadata will be added in the next stage.
6. **Output.** The predicted diagnosis, the probability of malignancy, and a *refer / monitor* recommendation (done; the threshold caught 95.4% of cancers on the test set). The Grad-CAM heatmap is the next stage.

3 Development Flow (How We Build It)

The project is carried out in seven phases (Fig. 2). Phases 1–4 are complete: the paper was written, both datasets were prepared, the three models were trained on an RTX 3060 laptop GPU (16–17 October 2026), and the measured results were added to the paper.

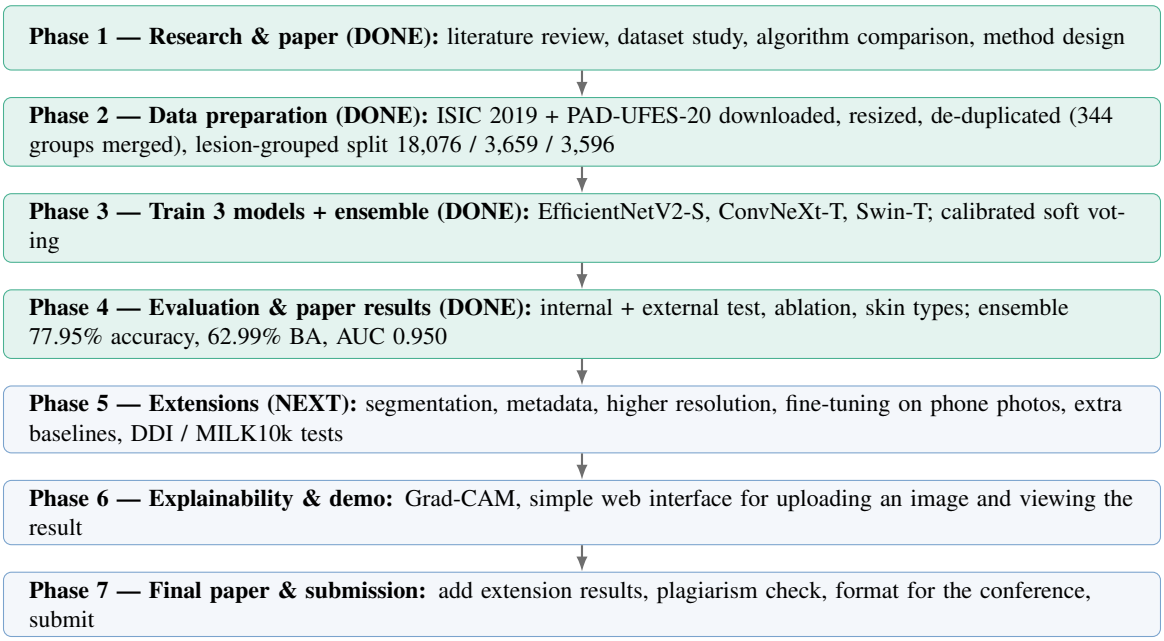

Figure 2: Development phases (green = completed).

3.1 Phase Details

Phase	Main tasks	Output	Status / time
1. Research & paper	Review 40+ studies; study 17 datasets; compare 11 algorithm families; design the method and evaluation protocol	IEEE paper, Datasets and Algorithms reports	Completed

Phase	Main tasks	Output	Status / time
2. Data preparation	Download ISIC 2019 (9.8 GB) and PAD-UFES-20 (3.6 GB); resize; image-hash de-duplication; map labels to 8 classes; lesion-grouped stratified split	Clean dataset, split files	Completed (about 1 hour incl. download)
3. Train 3 models + ensemble	EfficientNetV2-S, ConvNeXt-T, Swin-T with focal loss, AdamW, early stopping, TTA; temperature scaling; equal and learned soft voting	Trained models, predictions	Completed (43 + 49 + 36 min on RTX 3060)
4. Evaluation & paper results	Internal and external tests, ablation over model combinations, confusion matrix, ROC, bootstrap CIs, skin-type breakdown; results section, abstract, and conclusion of the paper	Training Results Report, updated paper	Completed
5. Extensions	Attention U-Net segmentation, metadata fusion, higher resolution (384 px), fine-tuning on phone photos, baselines (ResNet-50, DenseNet-121, MobileNetV2, DINOv2), DDI / MILK10k tests	Improved results	Next: about 3–4 weeks
6. Explainability & demo	Grad-CAM for each model; heatmap-in-lesion score; web demo (Streamlit or Flask)	Heatmap figures, demo app	About 2 weeks
7. Final paper	Add extension results, plagiarism check, format for the target conference, submit	Submission-ready paper	About 1 week

4 How the Paper Follows the Project

We wrote the paper *research-first*: the design and the test rules were fixed, with published evidence, before we saw any results. After training, the measured values were added in a new Experimental Results section (Section VI), so the paper now contains both the design and the real results. Table 2 shows how each section of the paper corresponds to the project.

Table 2: Paper sections mapped to project work.

Paper section	What it says	Project work	Status
Abstract & I. Introduction	The problem, our contributions, and the measured results	Motivation and objectives	Done (with real numbers)
II. Literature Review	40+ studies in 7 themes; Table I; six research gaps (G1–G6)	Phase 1 research	Done
III. Datasets	12 datasets reviewed; ISIC 2019 and PAD-UFES-20 used	Phase 2	Done
IV. Proposed Methodology	Framework (Fig. 3); implemented part: 3-model ensemble, focal loss, calibration; Algorithm 1; Table III	Phase 3; extensions in Phase 5	Core done
V. Experimental Design	Leakage-free protocol, metrics, fairness, ablation	Phase 4	Done
VI. Experimental Results	Training curves, internal/external results, confusion matrix, ROC, ablation, skin types (Figs. 4–7, Tables IV–VII)	Phase 4	Done
VII. Comparative Analysis	Published results of 11 algorithm families in 4 panels, plus our result	Phases 1 and 4	Done
VIII–IX. Discussion & Conclusion	Why the ensemble helps, limitations, accuracy and metrics, future work	Phase 4	Done

4.1 How Each Research Gap Becomes a Project Feature

Gap found in the literature	What our project did
G1: Results are not comparable (random splits, accuracy only)	Removed duplicates, split by lesion, reported balanced accuracy with confidence intervals
G2: No testing on outside data	Tested on PAD-UFES-20 (never used in training); MILK10k next
G3: Rare cancers (SCC, AK) are neglected	8-class training with focal loss; per-class recall reported
G4: Skin-tone bias	Results per Fitzpatrick type on PAD-UFES-20; DDI next
G5: Explanations not evaluated	Grad-CAM with heatmap-in-lesion score (Phase 6)
G6: Accurate models are too heavy	All 3 models trained on a 6 GB laptop GPU in under 1 hour each

5 Tools, Team Roles, and Deliverables

5.1 Tools and Technology

- **Language and framework:** Python, PyTorch, `timm` (pre-trained models), `segmentation_models_pytorch`.
- **Explainability:** `pytorch-grad-cam`.
- **Data:** pandas, OpenCV, `imagehash` (duplicate removal), scikit-learn (GroupKFold, metrics).
- **Hardware:** one NVIDIA GPU; we used an RTX 3060 Laptop GPU (6 GB). Google Colab or Kaggle GPUs also work.
- **Demo:** Streamlit or Flask web interface.
- **Paper:** LaTeX (IEEE template) on Overleaf.

5.2 Suggested Team Roles

Member	Responsibility
Member 1	Data pipeline: downloads, deduplication, label mapping, splits; segmentation model
Member 2	Classification: baselines, three backbones, metadata fusion, ensemble
Member 3	Evaluation: metrics, external and fairness tests, Grad-CAM, graphs; demo app
All + guide	Paper writing, review, and presentation

5.3 Deliverables

1. IEEE paper with measured results (complete).
2. Datasets, Algorithms, and Training Results reports (complete).
3. Source code with training and evaluation scripts (`SkinCancer-training/`, complete).
4. Trained models, results tables, and graphs (complete).
5. Web demo showing diagnosis, probability, heatmap, and referral (next).

5.4 Success Criteria and What We Achieved

Criterion	Result
Ensemble beats every single model	**Achieved:** 62.99% BA vs. 57.80–60.21%; 77.95% vs. 73.08–76.89% accuracy
Ensemble beats every pair of models (USP)	**Achieved:** 3 models $>$ any 2 models on all metrics
$\geq$95% malignant sensitivity at the referral threshold	**Achieved** internally (95.4% at 69.9% specificity); 78.9% on external phone photos
Internal-to-external drop measured and reported	**Achieved:** accuracy 77.95% $\rightarrow$ 37.25%; malignant AUC 0.934 $\rightarrow$ 0.873
Skin-tone gap measured	**Partly:** types I–II vs. III–IV similar; only 11 images of types V–VI (DDI needed)
Grad-CAM focuses on the lesion	**Next stage**

Summary for the guide: the project detects skin cancer from lesion images by combining two CNNs (EfficientNetV2-S, ConvNeXt-T) and one transformer (Swin-T). The paper was written first as the evidence-based design (literature review → gaps → datasets → method → test rules). The models were then trained on ISIC 2019 and tested on held-out and external data. The combined model reached **77.95% accuracy, 62.99% balanced accuracy, and AUC 0.950**, better than any single model, and these real results are now in the paper.